

Taste triggers a homeostatic temperature control in *Drosophila*

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Abstract

Summary

Hungry animals consistently show a desperate desire to obtain food. Even a brief sensory detection of food can trigger bursts of physiological and behavioral changes. However, the underlying mechanisms by which the sensation of food triggers the acute behavioral response remain elusive. We have previously shown in *Drosophila* that hunger drives a preference for low temperature. Because *Drosophila* is a small ectotherm, a preference for low temperature implies a low body temperature and a low metabolic rate. Here, we show that taste sensing triggers a switch from a low to a high temperature preference in hungry flies. We show that taste stimulation by artificial sweeteners or optogenetics triggers an acute warm preference, but is not sufficient to reach the fed state. Instead, nutrient intake is required to reach the fed state. The data suggest that starvation recovery is controlled by two components: taste-evoked and nutrient-induced warm preferences, and that taste and nutrient quality play distinct roles in starvation recovery. Animals are motivated to eat based on time of day or hunger. We found that clock genes and hunger signals profoundly control the taste-evoked warm preferences. Thus, our data suggest that the taste-evoked response is one of the critical layers of regulatory mechanisms representing internal energy homeostasis and metabolism.

eLife assessment

This paper presents **valuable** findings that gustation and nutrition might independently influence the preferred environmental temperature in flies. While some of the evidence is **solid**, support for other claims appears **incomplete** at this stage but can be addressed with some further experimentation. The finding that flies might thus exhibit a cephalic phase response similar to mammals will be of value for future investigations.

Introduction

Animals are constantly sensing environmental stimuli and changing their behavior or physiology based on their internal state ^{1–7}. Hungry animals are desperate for food. Immediately after seeing, smelling, or chewing food, even without absorbing nutrients, a burst of physiological changes is suddenly initiated in the body. These responses are known in mammals as the cephalic phase response (CPR). For example, a flood of saliva and gastrointestinal secretions prepares hungry animals to digest food ^{8–12}. Starvation results in lower body temperatures, and chewing food triggers a rapid increase in heat production, demonstrating CPR in thermogenesis ^{13–15}. Hungry animals are strongly attracted to food. However, the underlying mechanisms of how the sensation of food without nutrients triggers the behavioral response remain unclear.

To address this question, we used a relatively simple and versatile model organism, *Drosophila melanogaster*. Flies exhibit robust temperature preference behavior ^{16,17}. Due to the low mass of small ectotherms, their body temperatures are close to the ambient temperature ^{18,19}. Hunger stress forces flies to change their behavior and physiological response ^{20,21}. The hungry flies prefer a lower temperature ²², indicating lower body temperatures and lower metabolic rates ^{22–24}. Notably, this phenomenon of hunger-driven lower body temperature is similar to mammals, as the body temperature of starved animals is lower than that of fed animals ^{5–7}. Furthermore, flies exhibit robust feeding behavior ^{25,26}, and molecular and neural mechanisms of taste are well documented ^{2,27–32}. Therefore, we asked whether the taste cue triggers a robust behavioral recovery of temperature preference in starving flies.

Here we show in hungry flies that taste without nutrients induces a switch from a low to a high temperature preference. While taste leads to a warmer temperature preference, nutrient intake causes the flies to prefer an even warmer temperature. This nutrient-induced warm preference results in a complete recovery from starvation. Thus, taste-evoked warm preference is different from nutrient-induced warm preference and potentially similar physiology as CPR. Therefore, when animals emerge from starvation, they use a two-step approach to recovery: taste-evoked and nutrient-induced warm preference. While a rapid component is elicited by food taste alone, a slower component requires nutrient intake.

Animals are motivated to eat based on their internal state, such as time of day or degree of hunger. The circadian clock drives daily feeding rhythms ^{33–35} and anticipates meal timing. Daily feeding timing influences energy homeostasis and metabolism ^{36,37}. Hunger and satiety dramatically affect animal behavior and physiology ^{1–7} and control feeding. We found that clock genes and hunger signals are strongly required for taste-evoked warm preference. The data suggest that taste-evoked response is an indispensable physiological response that represents internal state. Taken together, our data shed new light on the role of the taste-evoked response and highlight a crucial aspect of our understanding of feeding state and energy homeostasis.

Results

Food detection triggers a warm preference

Flies exhibit robust temperature preference behavior ^{16,17}. To examine *Drosophila* temperature preference behavior, flies are released into a chamber with a temperature gradient of 16–34°C ³⁸ and subsequently accumulate at their preferred temperature (Tp) within 30 min. The *white*¹¹¹⁸ (*w*¹¹¹⁸) flies were fed fly food containing carbohydrate, protein and fat sources (Fig. 1A) and preferred 25.2±0.2 °C (Fig. 1B: fed, white bar). When *w*¹¹¹⁸ flies were starved

overnight with water only, they preferred 21.7 ± 0.3 °C (Fig. 1A, 1B: overnight starvation (STV), gray bar). Thus, starvation leads to a lower Tp. As we have previously reported, starvation strongly influences temperature preference²².

To examine how starved flies recover from lower Tp, they were offered fly food (Fig. 1A) and their Tp was tested. After 10 min, 30 min or 1 hr of fly food refeeding, starved flies preferred a temperature similar to that of fed flies (Fig. 1B: statistics shown as green stars, Table S1), suggesting a full recovery from starvation. On the other hand, refeeding after 5 min resulted in a warmer temperature than the starved flies. Nevertheless, Tp did not reach that of the fed flies (Fig. 1B). Therefore, 5 min fly food refeeding caused a partial recovery from the starved state (Fig. 1E, statistics shown as red stars, Table S1). Thus, our data suggest that food intake triggers a warm preference in starved flies.

Sucralose refeeding promotes a warm preference

Given that only 5 min refeeding caused hungry flies to prefer a warmer temperature (Fig. 1B), we hypothesized that food-sensing cues might be important for the warm preference. Sucralose is an artificial sweetener that activates sweet taste neurons in *Drosophila*³⁹ and modulates taste behaviors such as the proboscis extension reflex^{40–43}. Importantly, sucralose is a non-metabolizable sugar and has no calories. Therefore, to investigate how food-sensing cues are involved in warm preference, we examined how sucralose refeeding changes the temperature preference of starved flies. After starved flies were refeed sucralose for 10 min or 1 hr (Fig. 1A), they preferred a warmer temperature; however, Tp was halfway between Tps of fed and starved flies (Fig. 1C: blue, Table S1). Thus, sucralose ingestion induces a warm preference but shows a partial recovery of Tp from the starved state.

While refeeding the flies with food resulted in a full recovery of Tp from the starved state, refeeding them with sucralose resulted in only a partial recovery of Tp. Therefore, starved flies may use both taste cues and nutrients to fully recover Tp from the starved state. To evaluate this possibility, we used glucose, which contains both sweetness (gustatory cues) and nutrients (i.e., metabolizable sugar) (Fig. 1A), and tested glucose refeeding for 5 min, 10 min, and 1 hr. We found that 10 min or 1 hr glucose refeeding resulted in full recovery of Tp from the starved state (Fig. 1D and E: statistics shown in red NS, Table S1) and was significantly different from starved flies (Fig. 1D and E: statistics shown in green stars, Table S1). Thus, our data showed that sucralose refeeding induced partial recovery and glucose refeeding induced full recovery from the starved state.

However, it is still possible that the starved flies consumed glucose more or faster than sucralose during the first 10 min, which could result in a different warming preference. To rule out this possibility, we examined how often starved flies touched glucose, sucralose, or water during the 30 min using the Fly Liquid-food Interaction Counter (FLIC) system⁴⁴. The FLIC system assays allow us to monitor how much interaction between the fly and liquid food reflects feeding/drinking episodes. We found that starved flies touched glucose and sucralose food at similar frequencies and more frequently than water during the 30 min test period (Fig. 1F, Table S1). The data suggest that flies are likely to consume glucose and sucralose at similar rates and amounts. Therefore, we concluded that the differential effect of sucralose and glucose refeeding on temperature preference was not due to differences in feeding rate or amount. Instead, the gustatory cues likely induce partial Tp recovery, and the gustatory cue plus nutrients induce full Tp recovery.

Activation of sweet taste neurons leads to warm preference

We first focused on how taste elicits a warm preference. Sweet gustatory receptors (Grs) detect sweet taste. We used sweet Gr mutants and asked whether sweet Grs are involved in taste-evoked warm preference. Two different sweet Gr mutant strains, *Gr5a*^{-/-}; *Gr64a*^{-/-} and *Gr5a*^{-/-}; *Gr61a*^{-/-},

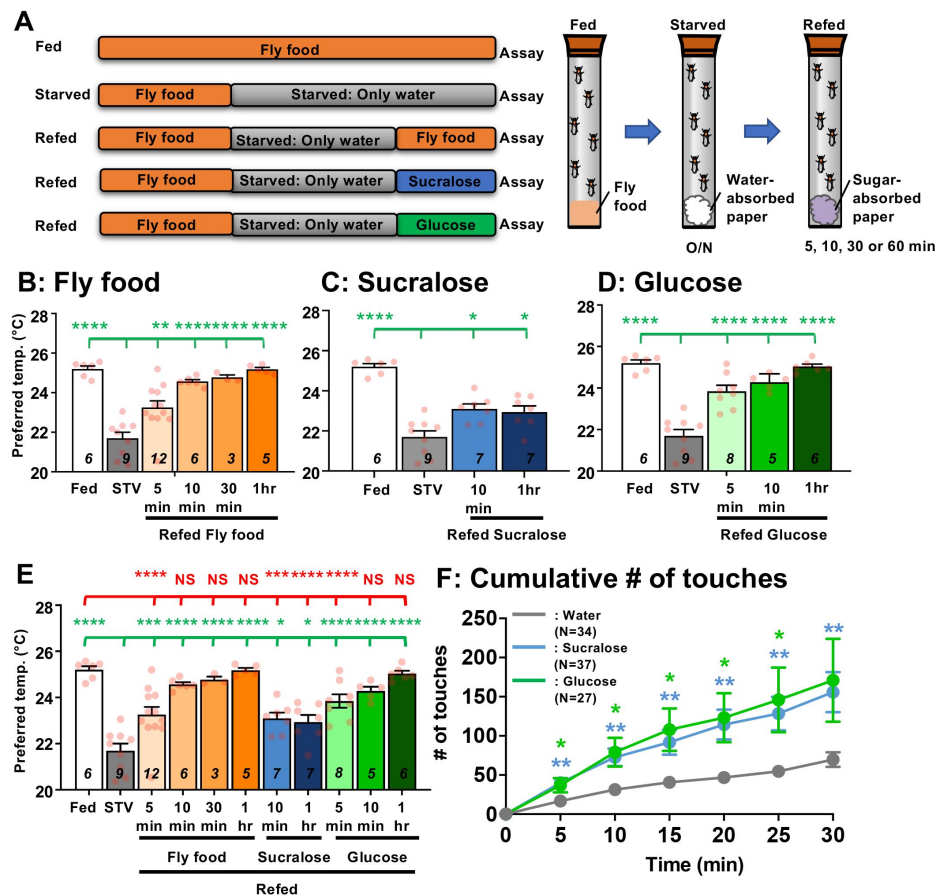


Figure 1.

Hungry flies switch from cold to warm preference upon food detection.

(A) A schematic diagram of the experimental conditions. Flies were reared on fly food. Emerging flies were kept on fly food for one or two days (orange) and then starved with water-absorbing paper for one or two nights (gray). Starved flies were re-fed with either fly food (orange), 2 mL sucralose (blue), or glucose solution (green) absorbed paper and then examined for temperature preference (Tp) behavioral assays. Details are provided in the experimental procedures.

(B-D) Comparison of Tp of *white*¹¹¹⁸ (*w*¹¹¹⁸) control flies between fed (white bar), starved (STV; gray bar), and re-fed (orange, blue, or green bar) states. Starved flies were re-fed with fly food (orange bar) for 5, 10, 30, or 60 min (1 hr) (B), 2.8 mM sucralose solution (blue bar) for 10 min or 1 hr (C), or 2.8 mM (equivalent to 5%) glucose solution (green bar) for 5 min, 10 min, or 1 hr (D). Behavioral experiments were performed at the specific time points, Zeitgeber time (ZT) 4-7. ZT0 and ZT12 are light on and light off, respectively. Dots on each bar indicate individual Tp in assays. Numbers in italics indicate the number of trials. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA was performed to compare Tp between each refeeding condition. Tukey's post hoc test was used to compare differences between experimental and starved conditions (green stars). Data are presented as mean Tp with SEM. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001. NS indicates no significance.

(E) Comparison of Tp among the following conditions in *w*¹¹¹⁸ (control) flies: fed (white bar), starved overnight (O/N) (water or STV; gray bar), re-fed with fly food (fly food; orange bar), sucralose (blue bars), or glucose (green bars). The same data are used in **Fig. 1B-D**. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA was performed to compare Tp between each refeeding condition. Red (or green) stars indicate Tukey's post hoc test comparing differences between experimental and fed (or starved) conditions, respectively. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001. NS indicates not significant.

(F) Feeding assay: The number of touches to water (gray), 2.8 mM (equivalent to 5%) glucose (green), or 2.8 mM sucralose in each solution (blue) was examined using *w*¹¹¹⁸ flies starved for 1 day. A cumulative number of touches to water or sugar solution for 0-30 min was plotted. Two-way ANOVA was used for multiple comparisons. Blue and green stars show Fisher's LSD post hoc test comparing sucralose (blue stars) or glucose (green stars) solution feeding to water drinking. All data shown are means with SEM. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001.

Gr64a-f^{-/-}, have reduced sugar sensitivity compared to the control ^{41,42,45,46}. We found that sweet Gr mutant flies exhibited a normal starvation response in which the Tp of starved flies was lower than that of fed flies (**Fig. 2A** and **B**, white and gray bars, Table S1). However, starved sweet Gr mutant flies did not increase Tp after 10 min sucralose refeeding (**Fig. 2A** and **B**, blue bars, Table S1). These data suggest that sweet Grs are involved in taste-evoked warm preference.

Sweet Grs are expressed in the sweet Gr-expressing neurons (GRNs) located in the proboscis and forelegs ^{46,47}. To determine whether sweet GRNs are involved in taste-evoked warm preference, we silenced all sweet GRNs. We expressed the inward rectifying K⁺ channel Kir2.1 (*uas-Kir*) ⁴⁸ with *Gr64f-Gal4*, which is expressed in all sweet GRNs in the proboscis and forelegs ^{41,46,47}. Knockdown of all sweet GRNs showed a normal starvation response (**Fig. 2C**, white and gray bars). However, flies silencing all sweet GRNs failed to show a warm preference after 10 min sucralose refeeding (**Fig. 2C**, blue bar, Table S1). This phenotype was similar to the data obtained with the sweet Gr mutant strains (**Fig. 2A** and **B**). On the other hand, control flies (*Gr64f-Gal4/+* and *uas-Kir/+*) showed a normal starvation response and a taste-evoked warm preference (**Fig. 2D** and **E**, gray and blue bars, respectively, Table S1). Thus, our data indicate that sweet GRNs are required for taste-evoked warm preference.

To further investigate whether activation of sweet GRNs induces a warm preference, we used the optogenetic approach, a red light sensitive channelrhodopsin, CsChrimson ^{49,50}. Starved flies were given water containing 0.8 mM all-trans-retinal (ATR), the chromophore required for CsChrimson activation. These flies were not fed sucralose; instead, gustatory neurons in starved flies were excited by red light pulses (flashing on and off at 10 Hz) for 10 min (**Fig. 2F**). In this case, although the flies were not refed, the gustatory neurons were artificially excited by CsChrimson activation so that we could evaluate the effect of excitation of sweet GRNs on taste-evoked warm preference.

CsChrimson was expressed in sweet GRNs in the proboscis and legs (all sweet GRNs) using *Gr64f-Gal4*. These flies showed a normal starvation response (**Fig. 2G-J**, white and gray bars, Table S1). Excitation of all sweet GRNs by red light pulses elicited a warm preference, and Tp was intermediate between fed and starved flies, suggesting partial recovery (**Fig. 2G**, yellow bar, Table S1). However, neither excitation of Gr5a- (**Fig. 2H**) nor Gr64a-expressing neurons (**Fig. 2I**) induced a warm preference (yellow bars, Table S1). While *Gr64a-Gal4* is expressed only in legs, *Gr5a-Gal4* is expressed in proboscis and legs, but does not cover all sweet GRNs like *Gr64f-Gal4* ^{46,47}. Notably, control flies (*UAS-CsChrimson/+*) did not show a warm preference to red light pulses with ATR application (**Fig. 2J**). Taken together, our data suggest that excitation of all sweet GRNs results in a warm preference.

The temperature-sensing neurons are involved in taste-evoked warm preference

The warm-sensing neurons, anterior cells (ACs) and the cold-sensing *R11F02-Gal4*-expressing neurons control temperature preference behavior ^{38,51,52}. Small ectotherms such as *Drosophila* set their Tp to avoid noxious temperatures using temperature information from cold- and warm-sensing neurons ^{16,17,38}. We have previously shown that starved flies choose a lower Tp, the so-called hunger-driven lower Tp ²². ACs control hunger-driven lower Tp but cold-sensing *R11F02-Gal4*-expressing neurons do not ²². ACs express transient receptor potential A1 (TrpA1), which responds to a warm temperature >25 °C ^{38,53}. The set point of ACs in fed flies, which is ~25 °C, is lowered in starved flies. Therefore, the lower set point of ACs corresponds to the lower Tp in starved flies.

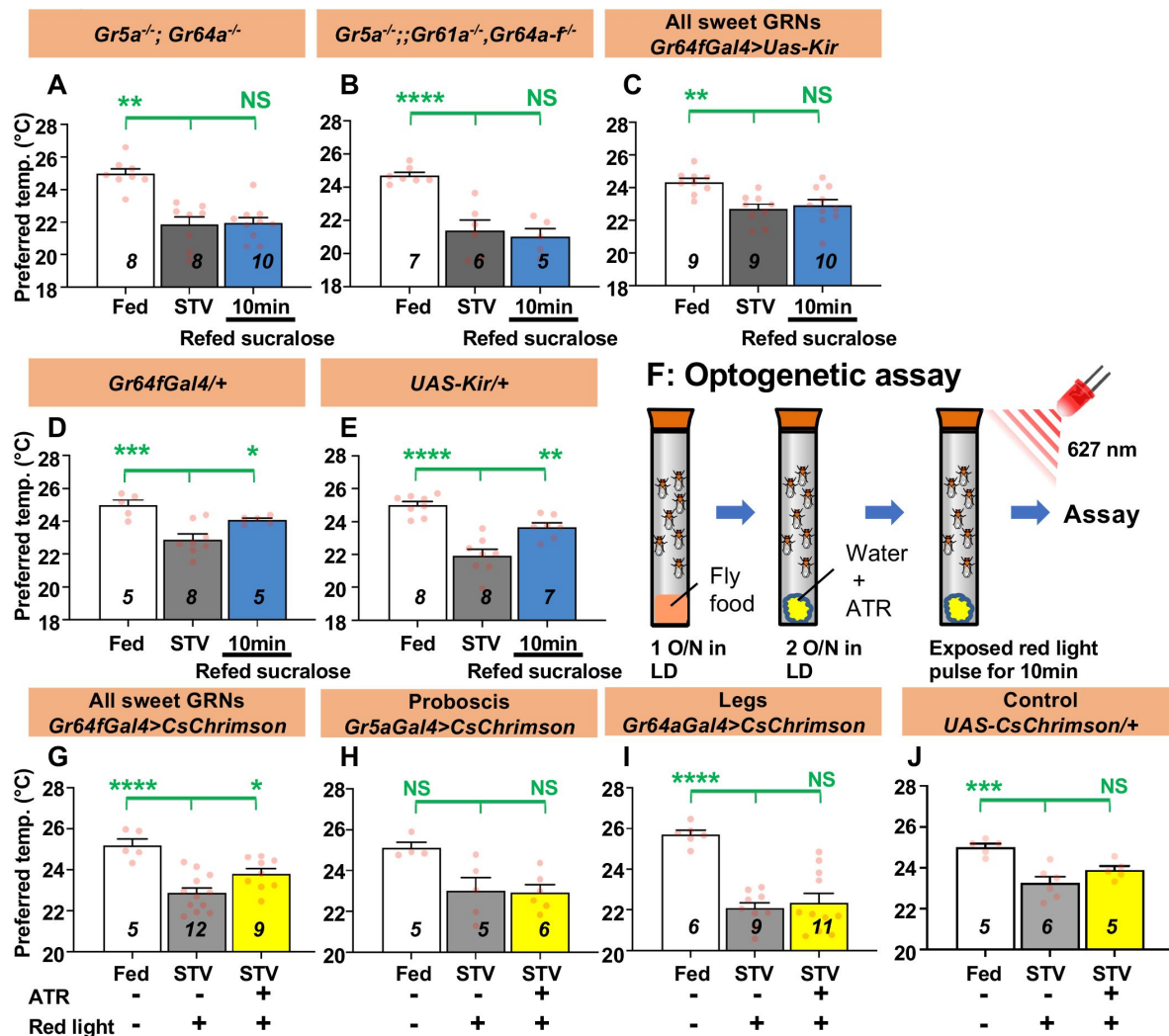


Figure 2.

Gustatory neurons are essential for taste-evoked warm preference.

(A-E) Comparison of Tp of flies between fed (white bar), starved (STV; gray bar), and refeed (blue bar) states. Starved flies were refeed with sucralose (blue bars) for 10 min. (F) Schematic of optogenetic activation assay.

(G-J) Comparison of Tp of flies between fed (white bar), starved (STV; gray bar), and starved with all-trans-retinal (ATR; yellow bars), which is the chromophore required for CsChrimson activation. Gustatory neurons in starved flies were excited by red light pulses (flashing on and off at 10 Hz) for 10 min. Behavioral experiments were conducted at ZT4-7. Dots on each bar indicate individual Tp in assays. Numbers in italics indicate the number of trials. Shapiro-Wilk test was performed for normality test. One-way ANOVA was used for statistical analysis. Green stars indicate Tukey's post hoc test compared between each experiment and the starved condition. Pooled data are shown in Fig. S3. All data shown are means with SEM. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. NS indicates not significant.

First, we asked whether ACs are involved in taste-evoked warm preference. Because the ACs are important for the hunger-driven lower Tp ²², the AC-silenced flies did not show a significant difference in Tp between fed and starved conditions for only one overnight starvation ²². Therefore, we first extended the starvation time to two nights so that the AC-silenced flies showed a significant difference in Tp between fed and starved conditions (**Fig. 3A**, white and gray bars, Table S1). Importantly, more prolonged starvation does not affect the ability of *w¹¹¹⁸* flies to recover (**Fig. S1**).

To examine whether ACs regulate taste-evoked warm preference, we refed sucralose to AC-silenced flies for 10 min. We found that the Tp of the refed flies was still similar to that of the starved flies (**Figure 3A**, blue bar, Table S1), indicating that sucralose refeeding could not restore Tp and that ACs are involved in taste-evoked warm preference. Significantly, even when the AC-silenced flies were starved for two nights, they were able to recover Tp to the normal food for 10 min and to glucose for 1 hr (**Fig. S3D**), suggesting that the starved AC-silenced flies were still capable of recovery.

Other temperature-sensing neurons involved in temperature preference behavior are cold-sensing *R11F02-Gal4*-expressing neurons ^{51,52}. To determine whether *R11F02-Gal4*-expressing neurons are involved in taste-evoked warm preference, we silenced *R11F02-Gal4*-expressing neurons using *uas-Kir*. We found that the flies showed a significant difference in Tp between fed and starved conditions, but the flies did not show a warm preference upon sucralose refeeding (**Fig. 3B**, gray and blue bars, Table S1). As controls, *TrpA1^{SH}-Gal4/+*, *R11F02-Gal4/+* and *uas-Kir/+* flies showed normal starvation response and taste-evoked warm preference (**Fig. 3C-E**).

To further ensure the results, we used optogenetics to artificially excite warm and cold neurons with *TrpA1^{SH}-Gal4* and *R11F02-Gal4*, respectively, by red light pulses for 10 min. We compared the Tp of starved flies with and without ATR under red light. Tp of starved flies with ATR was significantly increased (**Fig. 3F** and **G**, yellow bars, Table S1) compared to those without ATR (**Fig. 3F** and **G**, gray bars, Table S1). Therefore, these data indicate that ACs and *R11F02-Gal4*-expressing neurons are required for taste-evoked warm preference.

Internal state influences taste-evoked warm preference in hungry flies

Internal state strongly influences feeding motivation. However, how internal state influences starved animals to exhibit a food response remains unclear. Hunger represents the food-deficient state in the body, which induces the release of hunger signals such as neuropeptide Y (NPY). NPY promotes foraging and feeding behavior in mammals and flies ⁵⁴. While intracerebroventricular injection of NPY induces cephalic phase response (CPR) ⁵⁵, injection of NPY antagonists suppresses CPR in dogs, suggesting that NPY is a regulator of CPR in mammals ⁵⁶. Therefore, we first focused on neuropeptide F (NPF) and small neuropeptide F (sNPF), which are the *Drosophila* homologue and orthologue of mammalian NPY, respectively ⁵⁴, and asked whether they are involved in taste-evoked warm preference. In NPF mutant (*NPF^{-/-}*) or sNPF hypomorph (*sNPF hypo*) mutant, we found that the Tp of fed and starved flies were significantly different, showing a normal starved response (**Fig. 4A** and **B**: white vs. gray bars, Table S1). However, they failed to show a taste-evoked warm preference after 10 min sucralose refeeding (**Fig. 4A** and **B**: blue bars, Table S1). Thus, NPF and sNPF are required for taste-evoked warm preference after sucralose feeding.

Next, we asked whether NPF and sNPF are involved in the nutrient-induced warm preference during glucose refeeding. We found that starved *NPF^{-/-}* mutants significantly increased Tp after 1 hr glucose refeeding (**Fig. 4A**, green and dark green bars, Table S1). We also found that starved *sNPF hypo* mutants significantly increased Tp after 10 min and 1 hr of glucose refeeding (**Fig.**

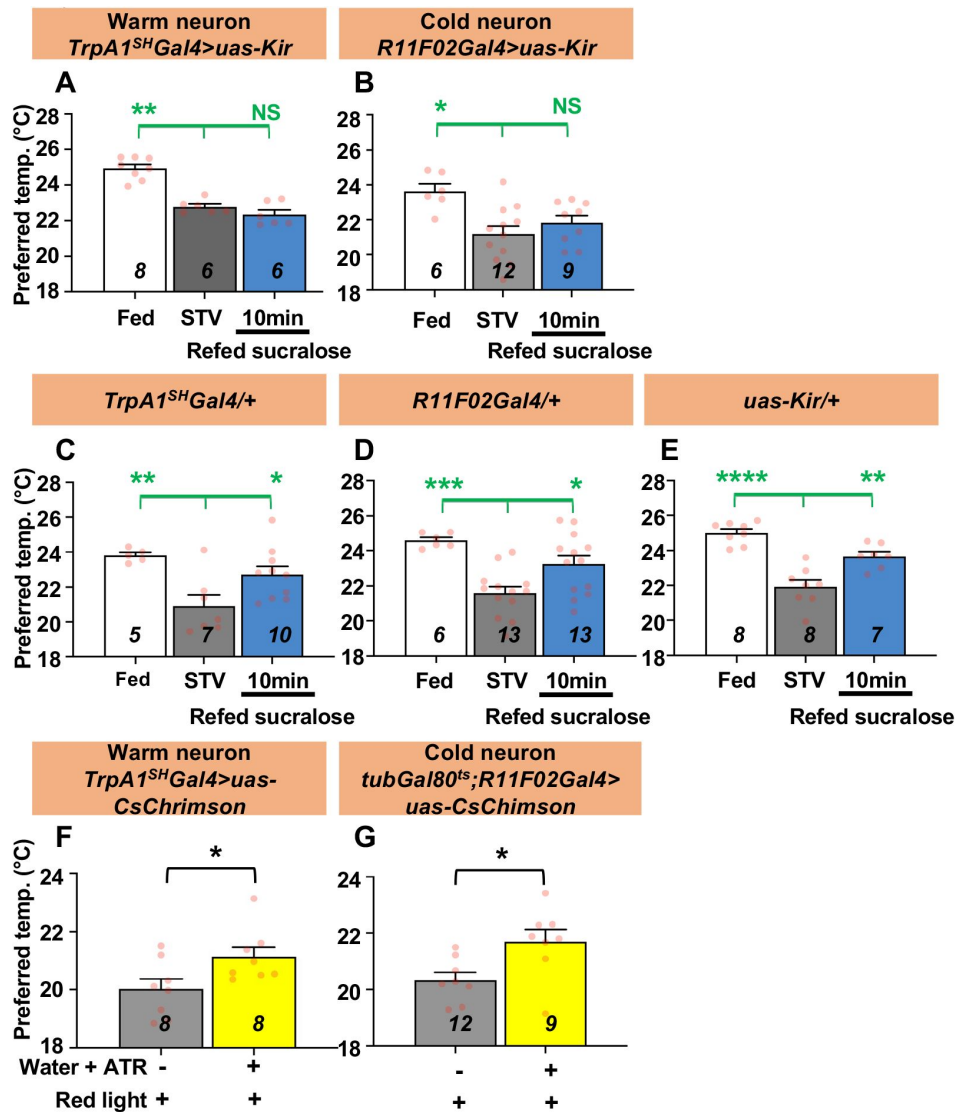


Figure 3.

Both warm- and cold-temperature-sensing neurons are involved with taste-evoked warm preference.

(A-E) Comparison of Tp of flies between fed (white bar), starved (STV; gray bar), and refed (blue bar) conditions. Starved flies were refed with sucralose (blue bars) for 10 min. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Green stars indicate Tukey's post hoc test or Dunn's test compared between each experiment and the starved condition. Pooled data are shown in Fig. S3. (F and G) Comparison of Tp between starved (STV; gray bar) and all-trans-retinal (ATR; yellow bars) starved flies. Warm neurons (F) or cold neurons (G) in starved flies expressed CsChrimson, which was excited by red light pulses for 10 min. The Shapiro-Wilk test was performed to test for normality. Student's t-test or Kolmogorov-Smirnov test was used for statistical analysis. (G) *tubGal80^{ts}; R11F02-Gal4>uas-CsChrimson* flies were reared at 18°C, and emerged adults were collected and stored at 29°C. See Materials and Methods for details. These behavioral experiments were conducted on ZT4-7. All data shown are means with SEM. *p < 0.05. **p < 0.01. ***p < 0.001. ****p < 0.0001. NS indicates no significance. The same data from *UAS-Kir/+* in Fig. 2E are used in Fig. 3E.

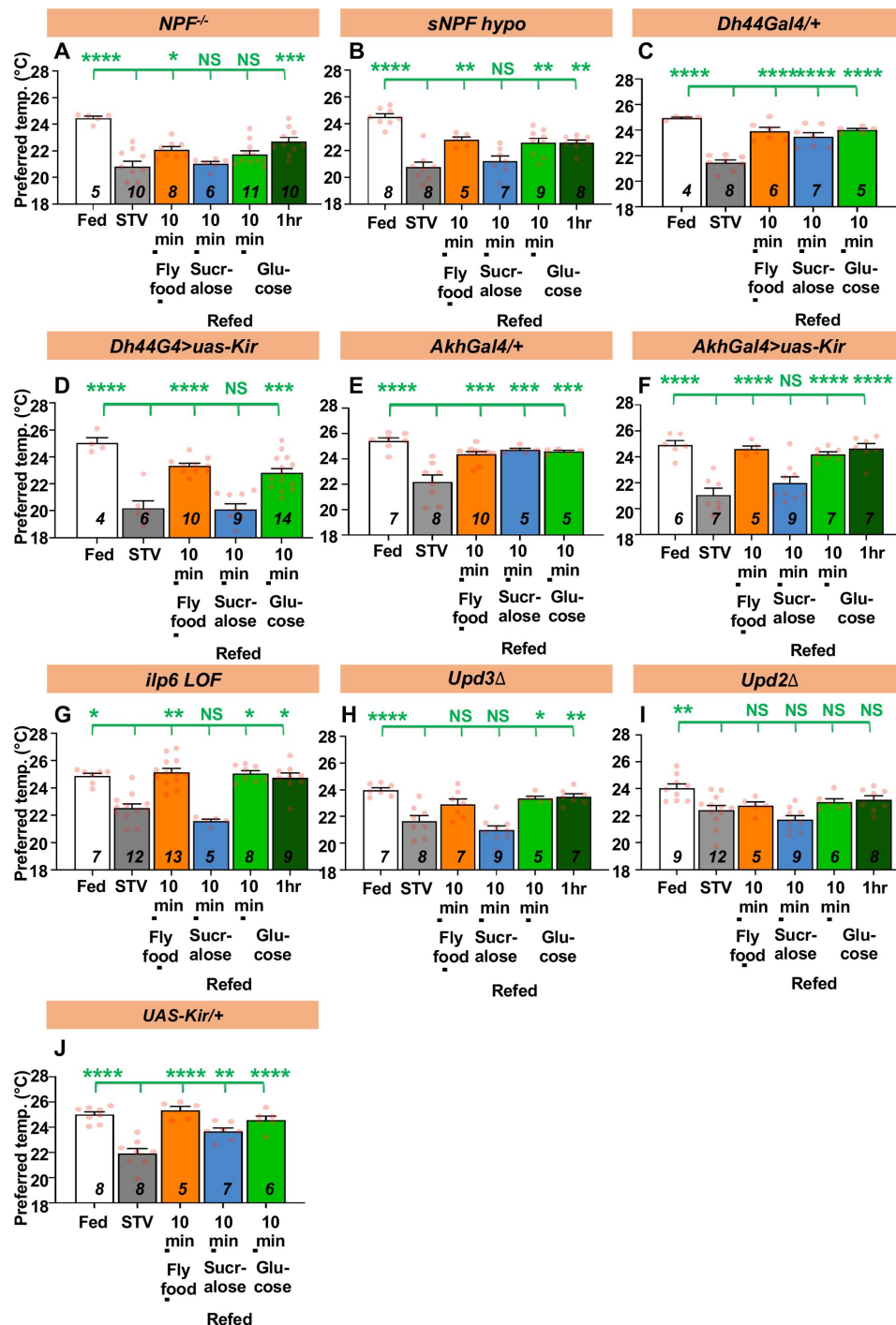


Figure 4.

Hunger signals are involved in taste-evoked warm preference.

Comparison of Tp of flies between fed (white bar), starved (STV; gray bar), and refed states (orange, blue, green, or dark green). Starved flies were refed with fly food (orange bar), sucralose (blue bar), or glucose (green bar) for 10 min. These behavioral experiments were conducted on ZT4-7. Dots on each bar indicate individual Tp in assays. Numbers in italics indicate the number of trials. Shapiro-Wilk test was performed for normality test. One-way ANOVA was used for statistical analysis. Green stars indicate Tukey's post hoc test compared between each experiment and the starved condition. Pooled data are shown in [Fig. S3](#). All data shown are means with SEM. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. NS indicates not significant.

4B [4B](#), green and dark green bars, Table S1). In addition, both *NPF*^{-/-} and *sNPF hypo* mutants increased Tp after 10 min fly food refeeding (Fig. 4A [4A](#) and B [4B](#), orange bars, Table S1). These data suggest that NPF and sNPF are unlikely to be critical for nutrient-induced warm preference.

Factors involving hunger regulate taste-evoked warm preference

Based on the above results, we hypothesized that hunger signals might be involved in taste-evoked warm preference. To test this hypothesis, we focused on several factors involved in the hunger state. The major hunger signals ²⁰[20](#), diuretic hormone 44 (DH44), and adipokinetic hormone (AKH) are the mammalian corticotropin-releasing hormone homologue ⁵⁷[57](#) and the functional glucagon homologue respectively ^{58,59}[58,59](#). We found that DH44- or AKH-neuron-silenced flies failed to show a taste-evoked warm preference after sucralose refeeding. However, they were able to increase Tp after 10 min glucose or fly food refeeding (Fig. 4D [4D](#) and F [4F](#), green and orange bars, Table S1), suggesting a normal nutrient-induced warm preference. The control flies (*Dh44-Gal4/+*, *Akh-Gal4/+*, and *UAS-Kir/+*) showed normal warm preference to sucralose (Fig. 4C, E [4C](#) and J [4J](#), blue bars, Table S1), glucose (Fig. 4C, E [4C](#) and J [4J](#), green bars, Table S1), and normal fly food refeeding (Fig. 4C, E [4C](#) and J [4J](#), orange bars, Table S1). The data suggest that DH44 or AKH neurons are required for taste-evoked warm preference, but not for nutrient-induced warm preference.

Insulin-like peptide 6 (Ilp6) is a homologue of mammalian insulin-like growth factor 1 (IGF1), and *ilp6* mRNA expression increases in starved flies ⁶⁰⁻⁶²[60-62](#). Because Ilp6 is important for the hunger-driven lower Tp ²²[22](#), the *ilp6* mutants did not show a significant difference in Tp between fed and starved conditions for only one overnight starvation ²²[22](#). Therefore, we first extended the starvation time to three nights. We found that *ilp6* loss-of-function (*ilp6 LOF*) mutant failed to show a taste-evoked warm preference after sucralose refeeding (Fig. 4G [4G](#), blue bars, Table S1), but showed a nutrient-induced warm preference after glucose refeeding (Fig. 4G [4G](#), green bars, Table S1). These data suggest that Ilp6 is required for taste-evoked warm preference but not for nutrient-induced warm preference after glucose or fly food refeeding.

Unpaired3 (Upd3) is a *Drosophila* cytokine that is upregulated under nutritional stress ⁶³[63](#). We found that *upd3* mutants failed to show a taste-evoked warm preference after sucralose refeeding (Fig. 4H [4H](#), blue bars, Table S1), but showed a nutrient-induced warm preference after glucose refeeding (Fig. 4G [4G](#) and H [4H](#), green bars, Table S1). These data suggest that Upd3 is required for taste-evoked warm preference, but not for nutrient-induced warm preference after glucose refeeding. Notably, *upd3* mutants failed to show nutrient-induced warm preference upon normal food feeding (Fig. 4H [4H](#), orange bars, Table S1).

We also examined the role of the satiety factor Unpaired2 (Upd2), a functional leptin homologue in flies, in taste-evoked warm preference ⁶⁴[64](#). *upd2* mutants failed to show a warm preference after refeeding of sucralose, glucose or fly food (Fig. 4I [4I](#), blue, green and orange bars, Table S1). Thus, our data suggest that factors involved in the hunger state are required for taste-evoked warm preference.

Flies show a taste-evoked warm preference at all times of the day

Animals anticipate feeding schedules at a time of day that is tightly controlled by the circadian clock ³⁷[37](#). Flies show a rhythmic feeding pattern: one peak in the morning ⁴⁴[44](#) or two peaks in the morning and evening ⁶⁵[65](#). Because food cues induce a warm preference, we wondered whether feeding rhythm and taste-evoked warm preference are coordinated. If so, they should show a parallel phenotype.

Since flies exhibit one of the circadian outputs, the temperature preference rhythm (TPR) ⁶⁶[66](#), Tp gradually increases during the day and peaks in the evening. First, we tested starvation responses at Zeitgeber time (ZT) 1-3, 4-6, 7-9, and 10-12 in light and dark (LD) conditions. We found that both

w¹¹¹⁸ and *yellow¹ white¹ (y¹w¹)* flies exhibited higher Tp in the fed state and lower Tp in the starved state at all times during the day (**Fig. 5A** and **B**; black vs. gray line; Table S1). Next, we re-fed sucralose to starved flies at all time points tested and examined taste-evoked warm preference. While starved *y¹w¹* flies showed a taste-evoked warm preference at all time points (**Fig. 5B1**, gray vs. blue line, Table S1), starved *w¹¹¹⁸* flies showed a significant taste-evoked warm preference at ZT 4-6 and 10-12 (**Fig. 5A1**, gray vs. blue line, Table S1).

Because starved *w¹¹¹⁸* flies showed an advanced phase shift of TPR with a peak at ZT 7-9 (**Fig. 5A**, gray), it is likely that the highest Tp simply masks the taste-evoked warm preference at ZT 7-9. We also focused on nutrient- (carbohydrate-) induced warm preference. Starved *w¹¹¹⁸* and *y¹w¹* flies successfully increased Tp after 10 min glucose refeeding (**Fig. 5A2** and **B2**, green line, Table S1). Glucose refeeding for 1 hr resulted in Tp similar to that of fed *w¹¹¹⁸* flies (**Fig. 5A3**, green line). Because the feeding rhythm peaks in the morning or morning/evening^{44,65}, our data suggest that the feeding rhythm and taste-evoked warm preference do not occur in parallel.

Circadian clock genes are required for taste-evoked warm preference, but not nutrient-induced warm preference

We asked whether the circadian clock is involved in taste-evoked warm preference. We used clock gene null mutants, *period⁰¹ (per⁰¹)* and *timeless⁰¹ (tim⁰¹)*. Although they showed significant starvation responses (**Fig. 5C** and **D**, black and gray lines, Table S1), neither starved *per⁰¹* nor *tim⁰¹* mutants could exhibit taste-evoked warm preference upon sucralose refeeding (**Fig. 5C1** and **D1**, blue lines, Table S1). Nevertheless, they fully recovered upon glucose refeeding in LD at any time of day (**Fig. 5C2** and **D2**, green lines, Table S1). Therefore, our data suggest that clock genes are required for taste-evoked warm preference, but not for nutrient-induced warm preference.

However, starved *per⁰¹* and *tim⁰¹* mutants may eat sucralose less frequently than glucose, which could result in a failure to show a taste-evoked warm preference. Therefore, we examined how often starved *per⁰¹* and *tim⁰¹* mutants touched glucose, sucralose, or water during the 30 min using FLIC assays⁴⁴ (**Fig. S2A-C**). Interestingly, starved *per⁰¹* and *tim⁰¹* mutants touched water significantly more often than glucose or sucralose (**Fig. S2B** and **C**, Table S1). Although starved *tim⁰¹* flies touched glucose slightly more than sucralose for only 10 min, this phenotype is not consistent with *per⁰¹* and *w¹¹¹⁸* flies. Thus, we conclude that in *per⁰¹* and *w¹¹¹⁸* flies, the differential response between taste-evoked and nutrient-induced warm preferences is not due to feeding rate or amount.

The sweet GRNs and temperature-sensing neurons contribute both taste-evoked and nutrient (carbohydrate)-induced warm preference

Our data suggest that taste-evoked and nutrient- (carbohydrate-) induced warm preferences are controlled separately. We examined how these differences are coordinated. We show that genes involved in internal state tend to control taste-evoked warm preferences. Since gustatory or temperature-sensitive neurons are required for taste-evoked warm preference (**Figs. 2** and **3**), we asked whether they contribute to nutrient-induced warm preference.

To this end, we used all-sweet-GRNs-silenced flies (*Gr64f-Gal4>uas-Kir*), two Grs mutant strains (*Gr5a^{-/-}; Gr64a^{-/-}* and *Gr5a^{-/-}; Gr61a^{-/-}, Gr64a^{-/-}*), and the warm- or cold-neuron-silenced flies (*TrpA1^{SH}-Gal4* or *R11F02-Gal4>uas-Kir*). We found that all these starved flies did not increase Tp after 10 min glucose intake (**Fig. S3A-E**, green bars, Table S1), but increased Tp after 10 min refeeding with fly food containing carbohydrate, fat, and protein (**Fig. S3A-E**, orange bars, Table S1). As an exception, AC-silenced flies increased Tp after 1 hr glucose intake (**Fig. S3D**, green bars, Table S1). All control flies showed normal responses to both 10 min glucose refeeding and fly

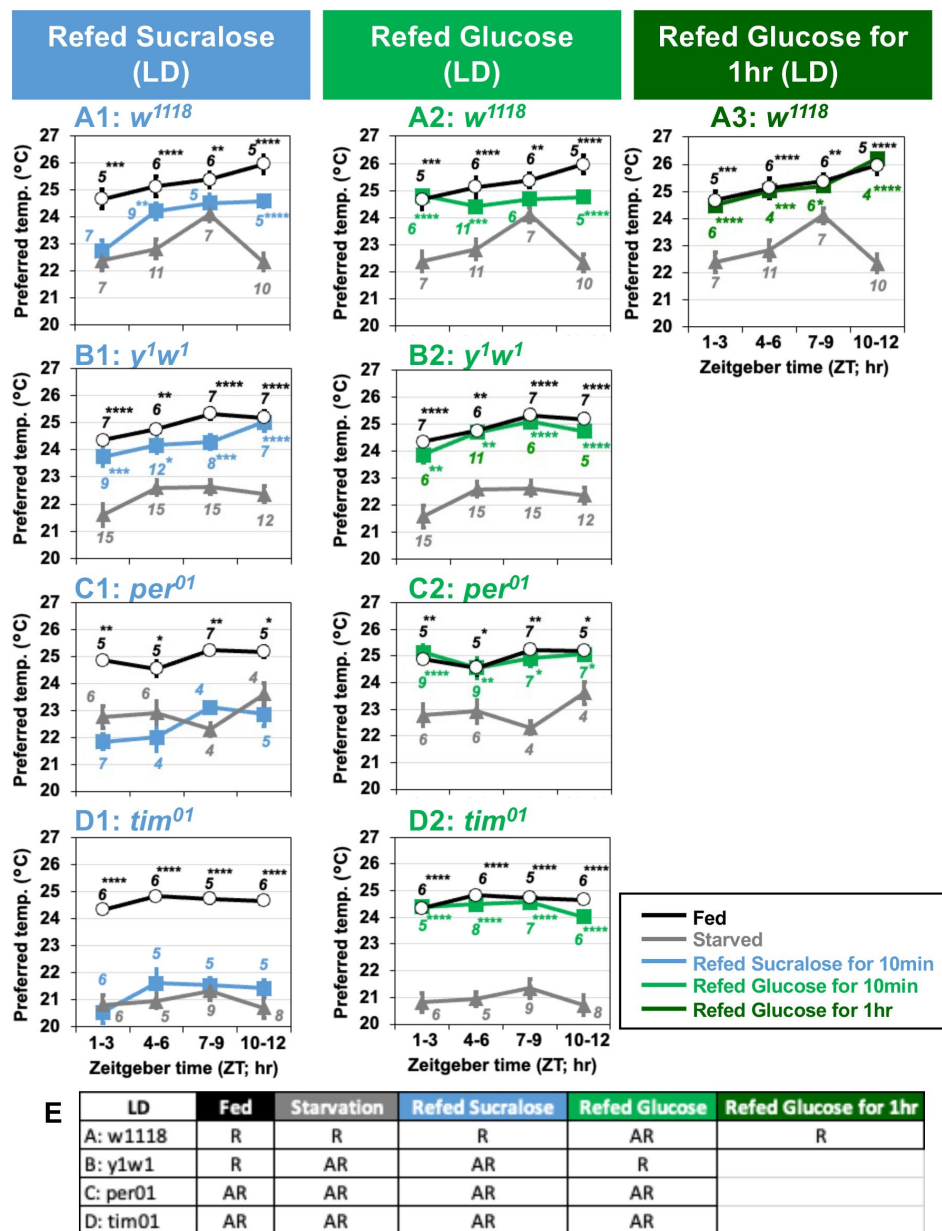


Figure 5.

Clock genes are involved in taste-evoked warm preference.

(A-D) Comparison of Tp of flies between fed (white circles), starved (STV; gray triangles), and refed (blue, green, or dark green squares) states. Flies were starved for 24 hrs. Starved flies were refed with sucralose (blue squares, A1-D1) or glucose for 10 min (green squares, A2-D2) or 1 hr (dark green squares, A3). After the 10 min or 1 hr refeeding, the temperature preference behavior assays were performed immediately at ZT1-3, 4-6, 7-9, and 10-12 in LD. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Stars indicate Tukey's post hoc test or Dunn's test compared between each experiment and starved condition at the same time point. All data shown are means with SEM. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$.

(E) Comparison of Tp during daytime in each feeding state. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis between ZT1-3 and ZT7-9 or ZT10-12. R and AR indicate rhythmic and arrhythmic, respectively, during the daytime.

food intake (**Fig. S3F-I**, green bars, Table S1). The data suggest that gustatory neurons and temperature-sensing neurons are required for warm preference in carbohydrate refeeding, but not in other foods such as fat or protein (see Discussion). Because flies have sensory neurons that detect fatty acids^{67–69} or amino acids^{70–73}, these neurons may drive the response to fly food intake. This is likely why the sweet- or temperature-insensitive flies can still recover after eating fly food (**Fig S3A-E**, orange bars).

Discussion

When animals are hungry, sensory detection of food (sight, smell, or chewing) initiates digestion even before the food enters the stomach. The food-evoked responses are also observed in thermogenesis, heart rate, and respiratory rate in mammals^{13,14,74}. These responses are referred to as the cephalic phase response (CPR) and contribute to the physiological regulation of digestion, nutrient homeostasis, and daily energy homeostasis¹⁰. While starved flies show a cold preference, we show here that the food cue, such as the excitation of gustatory neurons, triggers a warm preference, and the nutritional value triggers an even higher warm preference. Thus, when flies exit the starvation state, they use a two-step approach to recovery, taste-evoked and nutrient-induced warm preferences. The taste-evoked warm preference in *Drosophila* may be a physiological response potentially equivalent to CPR in mammals. Furthermore, we found that internal needs, controlled by hunger signals and circadian clock genes, influence taste-evoked warm preference. Thus, we propose that the taste-evoked response plays an important role in recovery and represents another layer of regulation of energy homeostasis.

Tp is determined by the taste cue

Starved flies increase Tp in response to a nutrient-free taste cue (**Fig. 1C** and **E**), resulting in a taste-evoked warm preference. We showed that silencing of ACs or cold neurons caused a loss of taste-evoked warm preference (**Figs. 3** and S3), and that excitation of ACs or cold neurons induced a taste-evoked warm preference (**Fig. 3**). The data suggest that both warm and cold neurons are important for taste-evoked warm preference: while ACs are required for the hunger-driven lower Tp²², both ACs and cold neurons are likely to be important for this taste-evoked warm preference.

The hunger-driven lower Tp is a slower response because starvation gradually lowers their Tp²². In contrast, the taste-evoked warm preference is a rapid response. Once ACs and cold neurons are directly or indirectly activated, starved flies quickly move to a warmer area. This is interesting because even when ACs and cold neurons are activated by warm and cold, respectively, the activation of these neurons causes a warm preference. Given that the sensory detection of food (sight, smell, or chewing food) triggers CPR, the activation of these sensory neurons may induce CPR. Tp in sucralose-fed hungry flies is between that of fed and starved flies (**Figs. 2**, **3**, and S3), making it difficult to detect the smaller temperature differences using the calcium imaging experiments. Therefore, we speculate that both ACs and cold neurons may facilitate rapid recovery from starvation so that flies can quickly return to their preferred temperature - body temperature - to a normal state.

Internal state influences taste-evoked warm preference

We show that mutants of genes involved in hunger and the circadian clock fail to show taste-evoked warm preference, suggesting that hunger and clock genes are important for taste-evoked warm preference. At a certain time of day, animals are hungry for food^{44,65}. Therefore, the "hungry" state acts as a gatekeeper, opening the gate of the circuits when hungry flies detect the food information that leads to taste-evoked warm preference (**Fig. S4**, blue arrow). While most hunger signals are important only for taste-evoked warm preference, some hunger signals are also

required for both taste-evoked and nutrient-induced warm preferences (Fig. 4). The data suggest that internal state is more likely to be associated with taste-evoked warm preference than nutrient-induced warm preference. It is noteworthy that the sensory signals contribute to both taste-evoked warm preference and nutrient-induced warm preferences (Figs. S1, S3 and S4, blue and green arrows). Therefore, taste-evoked warm preference and nutrient-induced warm preference are differentiated at the level of the internal state, but not at the sensory level. This idea is analogous to appetitive memory formation. Sweet taste and nutrients regulate the different layers of the memory formation process. Recent evidence suggests that the rewarding process can be subdivided; the sweet taste is for short-term memory and the nutrient is for long-term memory^{75–77}. The data suggest that the taste-evoked response functions differently from the nutrient-induced response. Therefore, taste sensation is not just the precursor to nutrient sensing/absorption, but plays an essential role in the rapid initiation of a taste-evoked behavior that would help the animal survive.

How do hunger signals or clock genes contribute to taste-evoked warm preference?

The hunger signaling hormones/peptides studied in this project are important for taste modulation. For example, mammalian NPY and its *Drosophila* homolog NPF modulate the output of taste signals^{43,78,79}. The AKH receptor is expressed in a subset of gustatory neurons that may modulate taste information for carbohydrate metabolism⁸⁰. Therefore, the hunger signals are likely to modulate the downstream of the sensory neurons, which may result in a taste-evoked warm preference.

Circadian clock genes control and coordinate the expression of many clock-controlled genes in the body^{81–84}. Therefore, we expect that the absence of clock genes will disrupt the molecular and neural networks of cellular homeostasis, including metabolism, that are essential for animal life. For example, taste neurons express clock genes, and impaired clock function in taste neurons disrupts daily rhythms in feeding behavior⁸⁵. Temperature-sensing neurons transmit hot or cold temperature information to central clock neurons^{86–90}. Therefore, the disrupted central clock in clock mutants may respond imprecisely to temperature signals. There are many possible reasons why the lack of clock gene expression in the brain is likely to cause abnormal taste-evoked warm preference.

In addition, hunger signals may contribute to the regulation of circadian output. DH44 is located in the dorsomedial region of the fly brain, the pars intercerebralis (PI), and DH44-expressing neurons play a role in the output pathway of the central clock^{91,92}. Insulin-producing cells (IPCs) are also located in addition to DH44-expressing neurons^{93–96}. IPCs receive a variety of information, including circadian^{92,97} and metabolic signals^{60–62,64}, and then transduce the signals downstream to release IIPs. Both Upd2 and Ilp6, which are expressed in the fat body respond to metabolic states and remotely regulate Ilp expression^{60–62,64}. Insect fat body is analogous to the fat tissues and liver in the vertebrates^{98,99}. Therefore, each hunger signal may have its specific function for taste-evoked warm preference. Further studies are needed to describe the entire process.

Taste-evoked warm preference may be CPR in flies

The introduction of food into the body disrupts the internal milieu, so CPR is a necessary process that helps animals prepare for digestion. Specifically, in mammals, taste leads to an immediate increase in body temperature and metabolic rate. Starvation results in lower body temperatures, and chewing food, even before it enters the stomach, triggers a rapid increase in heat production, demonstrating CPR in thermogenesis^{13,14}.

Starved flies have a lower T_p ²². Because *Drosophila* is a small ectotherm, the lower T_p indicates a lower body temperature ^{19,20}. Even when the flies do not receive food, the sweet taste and the excitation of sweet neurons induce starved flies to show a warm preference, which eventually leads to a warmer body temperature. Thus, the taste-evoked warm preference in *Drosophila* may be a physiological response equivalent to one of the CPRs observed in mammals. Thus, in both flies and mammals, starvation leads to lower body temperatures, and food cues quickly initiate a rise in body temperature before food enters their body.

CPR is essential because both starved mammals and starved flies must rapidly regulate their body temperature to survive

As soon as starving flies taste food, the sensory signals trigger CPR. They can move to a warmer place to prepare to raise their body temperature (Fig. S4, blue arrows). CPR may allow flies to choose a more hospitable place to restore their physiological state and allow for higher metabolism, and eventually they move to the next step, such as foraging and actively searching for a mate before competitors arrive. As such, CPR may be a strategy for the fly's survival. Similarly, starvation or malnutrition in mammals causes lower body temperatures ^{6,7,100}, and biting food triggers heat production, which is CPR ^{13–15}. Thus, while starvation causes lower body temperatures in both flies and mammals, food cues initiate CPR in body temperature elevation and nutrient intake, resulting in full recovery from the starved state. Our data suggest that *Drosophila* CPR may be a physiological response equivalent to CPR observed in other animals. Thus, *Drosophila* may shed new light on the regulation of CPR and provide a deeper understanding of the relationship between CPR and metabolism.

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Contributions

F.N.H and Y.U designed the experiments. Y.U, E.N., T.N. and J.S. performed the temperature preference behavioral assays and data analysis. Y.U., S.H.S., S.S.U. and J.C. performed the feeding experiments and the data analysis. F.N.H and Y.U wrote the manuscript.

Declaration of Interest

The authors declare no competing interests□

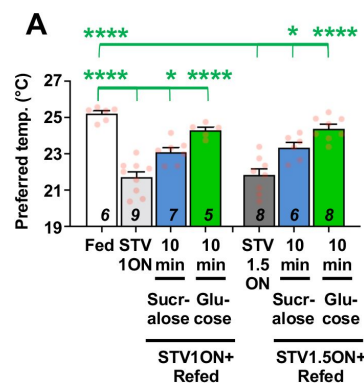


Figure S1.

The starvation period is unlikely to affect the ability to recover (Related to Fig. 3)

Comparison of T_p of w^{1118} control flies between fed (white bar), starved (STV; gray bar), and refed (STV; gray bar) and refed (blue or green bar). Flies were starved for overnight (STV1ON; gray bar) or 1.5 ON (STV1.5ON; dark gray bar), i.e. starved for 18-21 hrs. starved for 18-21 hrs or 26-29 hrs. Starved flies were refed with sucralose for 10 min (blue bar) or glucose for 10 min (green bar). These behavioral experiments were performed on ZT4-7. Dots on each bar indicate individual T_p in the assays. Numbers in italics indicate the number of trials. Per Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Green stars indicate Tukey's post hoc test or Dunn's test was used to compare between each experiment and starved condition, and starved condition. All data shown are means with sem. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. NS indicates no significance. The same data from STV1ON, STV1ON+refed sucralose for 10 min and STV1ON+refed sucralose for 10 min in Fig. 1C-E are used in Fig. S1.

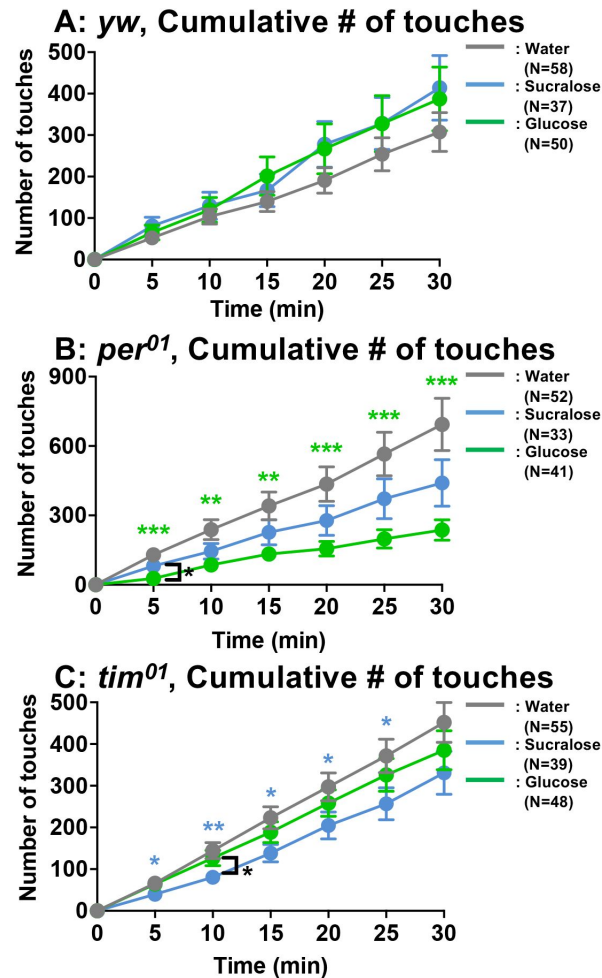


Figure S2.

Number of touches to water, sucralose and glucose using *yw*, *per*⁰¹ and *tim*⁰¹ flies. (Related to Figures 1, 5, and 6)

Feeding assay: The number of touches to water (gray), 2.8 mM (equivalent to 5%) glucose (green), or 2.8 mM sucralose solution (blue) was examined using *yw*, *per*⁰¹, or *tim*⁰¹ flies starved for 1 day. Cumulative number of touches for 0-30 min was plotted. Two-way ANOVA was used for multiple comparisons. Blue and green stars show Fisher's LSD post hoc test comparing sucralose (blue stars) or glucose (green stars) solution feeding to water drinking. Black stars show the statistical difference between sucralose and glucose refeeding. All data shown are means with SEM. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001.

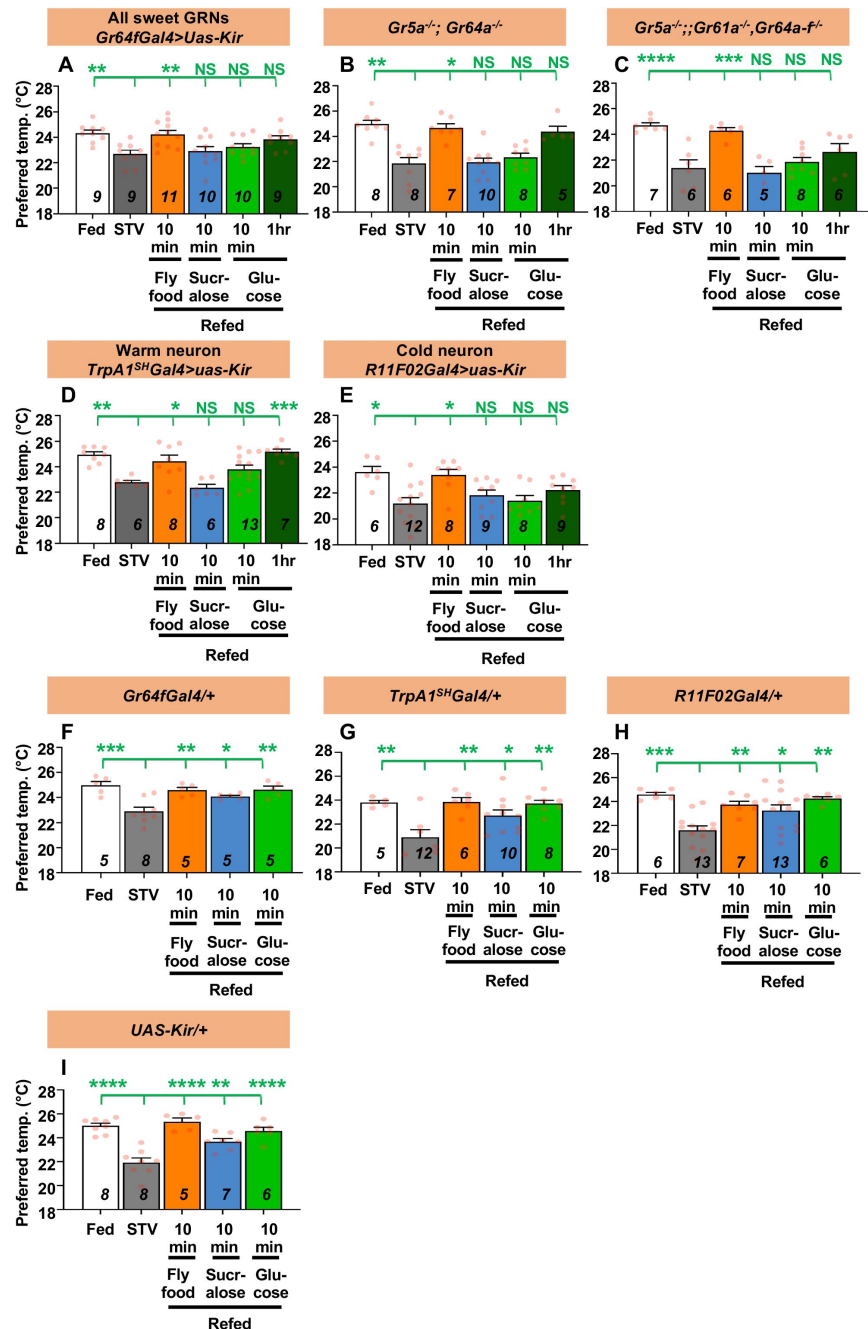


Figure S3.

Taste- or temperature-sensing neurons are not important for nutrient- (carbohydrate-) induced warm preference (Related to Figures 2 and 3).

Comparison of Tp of flies between fed (white bar), starved (STV; gray bar), and refed (orange, blue, green, or dark green bar) conditions. Starved flies were refed with fly food for 10 min (fly food; orange bar), sucralose for 10 min (blue bar), or glucose for 10 min (green bar) or 1 hr (dark green bar). These behavioral experiments were conducted on ZT4-7. The dots on each bar indicate individual Tp in the assays. Numbers in italics indicate the number of experiments. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Green stars indicate Tukey's post hoc test or Dunn's test compared between each experiment and the starved condition. All data shown are means with sem. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. NS indicates no significance. The same data from *UAS-Kir/+* in Fig. 2E are used in Fig. S3I.

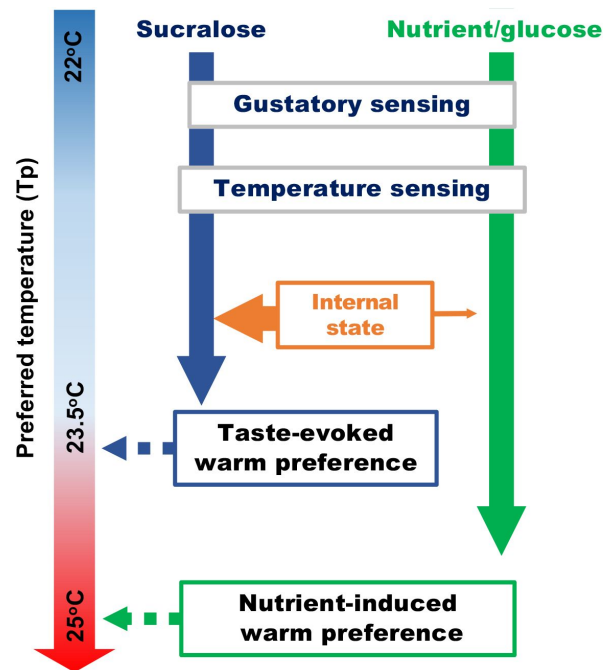


Figure S4.

Model

A schematic diagram of Tp recovery by taste-evoked warm preference (blue arrows) and nutrient-induced warm preference (green arrows).

Materials and Methods

All flies were reared under 12 hr-light/12 hr-dark (LD) cycles at 25°C and 60-70% humidity in an incubator (DRO33SD, Powers Scientific Inc.) with an electric timer (light on: 8am; light off: 8pm). The light intensity is 1000-1400 lux. All flies were reared on custom fly food recipe, with the following composition per 1 L of food: 6.0 g sucrose, 7.3 g agar, 44.6 g cornmeal, 22.3 g yeast and 16.3 mL molasses as described previously⁵². *white*¹¹¹⁸ (*w*¹¹¹⁸) and *yellow*¹ *white*¹ (*y*¹*w*¹) flies were used as control flies. *EP5Δ; Gr64a*¹ (*Gr5a*^{-/-}; *Gr64a*^{-/-}) was kindly provided by Dr. Anupama Dahanukar⁴¹, *R1; Gr5a-LexA*; +; *ΔGr61a, ΔGr64a-f* (*Gr5a*^{-/-}; *Gr61a*^{-/-}; *Gr64a-f*^{-/-}) were kindly provided by Dr. Hubert Amrein^{45,46}, *Dh44-Gal4* was kindly provided by Dr. Greg Suh⁴², *TrpA1*^{SH}-*gal4* was kindly provided by Dr. Paul A. Garrity³⁸. Other fly lines were provided by Bloomington Drosophila stock center and Vienna Drosophila Stock Center.

Temperature preference behavioral assay

The temperature preference behavior assays were examined using a gradient of temperatures, ranging from 16-34°C and were performed for 30 min in an environmental room maintained at 25°C /60%–70% humidity as described previously²². We prepared a total of 40-50 flies (male and female flies were mixed) for fed condition experiments and 90-100 flies for overnight(s) starved and refed condition experiments for one trial. Flies were never reused in subsequent trials. In starved and refed experiments, we prepared double the amount of flies required for one trial, as almost half of them died by starvation stress. Others climbed on the wall and ceiling (starved flies are usually hyperactive (Yang et al. 2015 PNAS)), even though we applied slippery coating chemicals (PTFE; Cat# 665800, Sigma or byFormica PTFE Plus, <https://byformica.com/products/fluon-plus-ptfe-escape-prevention-coating>) to the plexiglass covers.

The behavioral assays were performed for 30 min at Zeitgeber time (ZT) 4-7 (Light on and light off are defined as ZT0 and ZT12, respectively), and starvation was initiated at ZT9-10 (starved for 1, 2 or 3 O/N are 18-21 hrs, 42-45 hrs or 66-69 hrs, respectively). As for STV1.5 for Fig. S1, starvation was initiated at ZT1-2 (26-29 hrs). For the starvation assays, the collected flies were maintained with our fly food for at least one day for their maturation and subsequently moved into plastic vials with 3 mL of distilled water which is absorbed by a Kimwipe paper. For the refeeding experiments, starved flies were transferred into plastic vials with 2 mL of 2.8 mM sugar solution (Sucralose water/ Glucose water) for a certain time period. Sugar solution is absorbed by the half size of a Kimwipe.

After the 30-min behavioral assay, the number of flies whose bodies were completely located on the apparatus were counted. Flies whose bodies were partially or completely located on the walls of the apparatus cover were not included in the data analysis. The percentage of flies within each one-degree temperature interval on the apparatus was calculated by dividing the number of flies within each one-degree interval by the total number of flies on the apparatus. The location of each one-degree interval was determined by measuring the temperature at 6 different points on the bottom of the apparatus. Data points were plotted as the percentage of flies within a one-degree temperature interval. The weighted mean of *T_p* was calculated by totaling the product of the percentage of flies within a one-degree temperature interval and the corresponding temperature (e.g., fractional number of flies x 15.5°C + fractional number of flies x 16.5°C + fractional number of flies x 34.5°C). If the SEM of the averaged *T_p* was not < 0.3 after the five trials, we performed additional trials around 10 times until the SEM reached < 0.3.

Temperature preference rhythm (TPR) assay

For the TPR assays, we performed temperature preference behavior assays in different time windows during the day (zeitgeber time (ZT) or circadian time (CT) 1-3, 4-6, 7-9, and 10-12) as described previously^{66,101}. Because starvation duration directly affects flies' *T_p*²²,

starvation was initiated at each time window to adjust the starvation duration at each time point, which means flies were starved for 24 hrs or 48 hrs but not 1 or 2 O/N. Each behavioral assay was not examined during these time periods (ZT or CT0-1 and 11.5-13) because of large phenotype variation around light on and light off.

Optogenetic activation

For the optogenetic activation of the target neurons for behavioral assays, red-light-sensitive channelrhodopsin, *UAS-CsChrimson*, was crossed with each Gal4 drivers. Flies were reared on fly food at 25°C and 60-70% humidity under LD cycles in an incubator (DRoS33SD, Powers Scientific Inc.) with an electric timer. After flies emerged, we collected adult flies and kept them with fly food for 1-2 days. Next day, flies were transferred into water with or without 0.8 mM all-trans retinal (ATR; #R2500, Sigma) diluted in dimethyl sulfoxide (DMSO; #472301, Sigma) for 2 O/N. To activate flies expressing *UAS-CsChrimson* crossed with Gal4 drivers, we employed 627 nm red-colored light-emitted diode (LED) equipped with pulse photoillumination system (10 Hz, 0.08 mWmm⁻²). Flies were irradiated with pulsed red light for 10 min, which is equivalent to refeeding time period. This photoillumination system was used in an incubator (Sanyo Scientific, MIR-154), and followed by temperature preference behavioral assays.

Optogenetic assay for *tubGal80^{ts}*; *R11F02-Gal4>uas-CsChrimson*

Because *R11F02-Gal4>uas-CsChrimson* flies died during the pupal stage, Gal4/Gal80^{ts} system was used to restrict *uas-CsChrimson* expression to the adult stage for optogenetic assays against the cold-sensing neurons (*R11F02-Gal4*-expressing neurons). Gal80^{ts} is a temperature sensitive allele of Gal80, which binds Gal4 and inhibits *CsChrimson* gene expression at 18°C. However, Gal80 is degraded at 29°C; in turn, the Gal4 activates *CsChrimson* gene expression (Southall et al. 2008 CSH protocol). *tubGal80^{ts}*; *R11F02-Gal4>uas-CsChrimson* flies were reared with fly food at 18°C. Emerged adult flies were collected and kept with fly food at 29°C. Next day, flies were transferred to water condition (starved condition) with or without 0.8 mM ATR for 2 O/N. Starved flies with or without ATR application were exposed to pulsed red light for 10 min (equivalent to refeeding period) and then immediately loaded in the behavior apparatus for behavioral assays to measure their Tp.

Feeding assay

To measure single fly feeding, we used *Drosophila* Fly Liquid Interaction Counter (FLIC) system (Ro et al. 2013 Plos one; Sable systems International <https://www.sablesys.com>). Groups of 1-2-day-old male and female flies were starved for 24 hrs, which was started from between ZT4-5 (12-1pm). Individual flies were then loaded into FLIC monitors. Flies were acclimatized to the monitors for 30 min, with access to water provided in the food wells. To start the feeding assay, water was replaced with 2.8 mM Sucralose or 2.8 mM Glucose solution (equivalent to 5% Glucose conc.) and the number of licks (touches) were recorded for 30 min. Assays were performed at ZT4-7. To account for potential confounding effect of startle response when changing the food, wells where water was replaced with new water were used as a control. FLIC raw data was analyzed using the FLIC R code master (Pletcher Lab, available at https://github.com/PletcherLab/FLIC_R_Code). The numbers of licks were obtained and summed into 5-min windows bin, while the cumulative licks were obtained by the consecutive sum of the licks in these bins.

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Editors

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Reviewer #1 (Public Review):

Summary:

This paper presents valuable findings that gustation and feeding state influence the preferred environmental temperature preference in flies. Interestingly, the authors showed that by refeeding starved animals with the non-nutritive sugar sucralose, they are able to tune their preference towards a higher temperature in addition to nutrient-dependent warm preference. The authors show that temperature-sensing and sweet-sensing gustatory neurons (SGNs) are involved in the former but not the latter. In addition, their data indicate that peptidergic signals involved in internal state and clock genes are required for taste-dependent warm preference behavior.

The authors made an analogy of their results to the cephalic phase response (CPR) in mammals where the thought, sight, and taste of food prepare the animal for the consumption of food and nutrients. They further linked this behavior to core regulatory genes and peptides controlling hunger and sleep in flies having homologues in mammals. These valuable behavioral results can be further investigated in flies with the advantage of being able to dissect the neural circuitry underlying CPR and nutrient homeostasis.

Strengths:

(1) The authors convincingly showed that tasting is sufficient to drive warm temperature preference behavior in starved flies and that it is independent of nutrient-driven warm preference.

(2) By using the genetic manipulation of key internal sensors and genes controlling internal feeding and sleep states such as DH44 neurons and the *per* genes for example, the authors linked gustation and temperature preference behavior control to the internal state of the animal.

Weaknesses:

(1) The title is somewhat misleading, as the term homeostatic temperature control linked to gustation only applies to starved flies.

(2) The authors used a temperature preference assay and refeeding for 5 minutes, 10 minutes, and 1 hour. Experimentally, it makes a difference if the flies are tested immediately after 10 minutes or at the same time point as flies allowed to feed for 1 hour. Is 10 minutes enough to change the internal state in a nutrition-dependent manner? Some of the authors' data hint at it (e.g. refeeding with fly food for 10 minutes), but it might be relevant to feed for 5/10 minutes and wait for 55/50min to do the assays at comparable time points.

(3) A figure depicting the temperature preference assay in Figure 1 would help illustrate the experimental approach. It is also not clear why Figure 1E is shown instead of full statistics on the individual panels shown above (the data is the same).

(4) The authors state that feeding rate and amount were not changed with sucralose and glucose. However, the FLIC assay they employed does not measure consumption, so this statement is not correct, and it is unclear if the intake of sucralose and glucose is indeed comparable. This limits some of the conclusions.

(5) The authors make a distinction between taste-induced and nutrient-induced warm preference. Yet the statistics in most figures only show the significance between the starved and re-fed flies, not the fed controls. As the recovery is in many cases incomplete and used as

a distinction of nutritive vs non-nutritive signals (see Figure 1E) it will be important to also show these additional statistics to allow conclusions about how complete the recovery is.

(6) The starvation period used is ranging from 1 to 3 days, as in some cases no effect was seen upon 1 day of starvation (e.g. with clock genes or temperature sensing neurons). While the authors do provide a comparison between 18-21 and 26-29 hours old flies in Figure S1, a comparison for 42-49 and 66-69 hours of starvation is missing. This also limits the conclusion as the "state" of the animal is likely quite different after 1 day vs. 3 days of starvation and, as stated by the authors, many flies die under these conditions.

(7) In Figure 2, glucose-induced refeeding was not tested in Gr mutants or silenced animals, which would hint at post-ingestive recovery mechanisms related to nutritional intake. This is only shown later (in Figure S3) but I think it would be more fitting to address this point here. The data presented in Figure S3 regarding the taste-evoked vs nutrient-dependent warm preference is quite important while in some parts preliminary. It would nonetheless be justified to put this data in the main figures. However, some of the conclusions here are not fully supported, in part due to different and low n numbers, which due to the inherent variability of the behavior do not allow statistically sound conclusions. The authors claim that sweet GRNs are only involved in taste-induced warm preference, however, glucose is also nutritive but, in several cases, does not rescue warm preference at all upon removal of GRN function (see Figures S3A-C). This indicates that the Gal4 lines and also the involved GRs are potentially expressed in tissues/neurons required for internal nutrient sensing.

(8) In Figure 4, fly food and glucose refeeding do not fully recover temperature preference after refeeding. With the statistical comparison to the fed control missing, this result is not consistent with the statement made in line 252. I feel this is an important point to distinguish between state-dependent and taste/nutrition-dependent changes.

(9) The conclusion that clock genes are required for taste-evoked warm preference is limited by the observation that they ingest less sucralose. In addition, the FLIC assay does not allow conclusions about the feeding amount, only the number of food interactions. Therefore, I think these results do not allow clear-cut conclusions about the impact of clock genes in this assay.

(10) CPR is known to be influenced by taste, thought, smell, and sight of food. As the discussion focused extensively on the CPR link to flies it would be interesting to find out whether the smell and sight of food also influence temperature preference behavior in animals with different feeding states.

(11) In the discussion in line 410ff the authors claim that "internal state is more likely to be associated with taste-evoked warm preference than nutrient-induced warm preference." This statement is not clear to me, as neuropeptides are involved in mediating internal state signals, both in the brain itself as well as from gut to brain. Thus, neuropeptidergic signals are also involved in nutrient-dependent state changes, the authors might just not have identified the peptides involved here. The global and developmental removal of these signals also limits the conclusions that can be drawn from the experiments, as many of these signals affect different states, circuits, and developmental progression.

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Reviewer #2 (Public Review):

Animals constantly adjust their behavior and physiology based on internal states. Hungry animals, desperate for food, exhibit physiological changes immediately upon sensing, smelling, or chewing food, known as the cephalic phase response (CPR), involving processes

like increased saliva and gastrointestinal secretions. While starvation lowers body temperature, the mechanisms underlying how the sensation of food without nutrients induces behavioral responses remain unclear. Hunger stress induces changes in both behavior and physiological responses, which in flies (or at least in *Drosophila melanogaster*) leads to a preference for lower temperatures, analogous to the hunger-driven lower body temperature observed in mammals. In this manuscript, the authors have used *Drosophila melanogaster* to investigate the issue of whether taste cues can robustly trigger behavioral recovery of temperature preference in starving animals. The authors find that food detection triggers a warm preference in flies. Starved flies recover their temperature preference after food intake, with a distinction between partial and full recovery based on the duration of refeeding. Sucralose, an artificial sweetener, induces a warm preference, suggesting the importance of food-sensing cues. The paper compares the effects of sucralose and glucose refeeding, indicating that both taste cues and nutrients contribute to temperature preference recovery. The authors show that sweet gustatory receptors (Grs) and sweet GRNs (Gustatory Receptor Neurons) play a crucial role in taste-evoked warm preference. Optogenetic experiments with CsChrimson support the idea that the excitation of sweet GRNs leads to a warm preference. The authors then examine the internal state's influence on taste-evoked warm preference, focusing on neuropeptide F (NPF) and small neuropeptide F (sNPF), analogous to mammalian neuropeptide Y. Mutations in NPF and sNPF result in a failure to exhibit taste-evoked warm preference, emphasizing their role in this process. However, these neuropeptides appear not to be critical for nutrient-induced warm preference, as indicated by increased temperature preference during glucose and fly food refeeding in mutant flies. The authors also explore the role of hunger-related factors in regulating taste-evoked warm preference. Hunger signals, including diuretic hormone (DH44) and adipokinetic hormone (AKH) neurons, are found to be essential for taste-evoked warm preference but not for nutrient-induced warm preference. Additionally, insulin-like peptides 6 (Ilp6) and Unpaired3 (Upd3), related to nutritional stress, are identified as crucial for taste-evoked warm preference. The investigation then extends into circadian rhythms, revealing that taste-evoked warm preference does not align with the feeding rhythm. While flies exhibit a rhythmic feeding pattern, taste-evoked warm preference occurs consistently, suggesting a lack of parallel coordination. Clock genes, crucial for circadian rhythms, are found to be necessary for taste-evoked warm preference but not for nutrient-induced warm preference.

Strengths:

A well-written and interesting study, investigating an intriguing issue. The claims, none of which to the best of my knowledge controversial, are backed by a substantial number of experiments.

Weakness:

The experimental setup used and the procedures for assessing the temperature preferences of flies are rather sparingly described. Additional details and data presentation would enhance the clarity and replicability of the study. I kindly request the authors to consider the following points: i) A schematic drawing or diagram illustrating the experimental setup for the temperature preference assay would greatly aid readers in understanding the spatial arrangement of the apparatus, temperature points, and the positioning of flies during the assay. The drawing should also be accompanied by specific details about the setup (dimensions, material, etc). ii) It would be beneficial to include a visual representation of the distribution of flies within the temperature gradient on the apparatus. A graphical representation, such as a heatmaps or histograms, showing the percentage of flies within each one-degree temperature bin, would offer insights into the preferences and behaviors of the flies during the assay. In addition to the detailed description of the assay and data analysis, the inclusion of actual data plots, especially for key findings or representative trials, would provide readers with a more direct visualization of the experimental outcomes. These additions will not only enhance the clarity of the presented information but also provide the reader with a more comprehensive understanding of the experimental setup and results. I

appreciate the authors' attention to these points and look forward to the potential inclusion of these elements in the revised manuscript.

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Reviewer #3 (Public Review):

Summary:

The manuscript by Yujiro Umezaki and colleagues aims to describe how taste stimuli influence temperature preference in *Drosophila*. Under starvation flies display a strong preference for cooler temperatures than under fed conditions that can be reversed by refeeding, demonstrating the strong impact of metabolism on temperature preference. In their present study, Umezaki and colleagues observed that such changes in temperature preference are not solely triggered by the metabolic state of the animal but that gustatory circuits and peptidergic signalling play a pivotal role in gustation-evoked alteration in temperature preference.

The study of Umezaki is definitively interesting and the findings in this manuscript will be of interest to a broad readership.

Strengths:

The authors demonstrate interesting new data on how taste input can influence temperature preference during starvation. They propose how gustatory pathways may work together with thermosensitive neurons, peptidergic neurons and finally try to bridge the gap between these neurons and clock genes. The study is very interesting and the data for each experiment alone are very convincing.

Weaknesses:

In my opinion, the authors have opened many new questions but did not fully answer the initial question - how do taste-sensing neurons influence temperature preferences? What are the mechanisms underlying this observation? Instead of jumping from gustatory neurons to thermosensitive neurons to peptidergic neurons to clock genes, the authors should have stayed within the one question they were asking at the beginning. How does sugar sensing influence the physiology of thermos-sensation in order to change temperature preference? Before addressing all the following questions of the manuscript the authors should first directly decipher the neuronal interplay between these two types of neurons.

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